

Sound

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What is sound?

Key takeaway:

- Sound takes up space

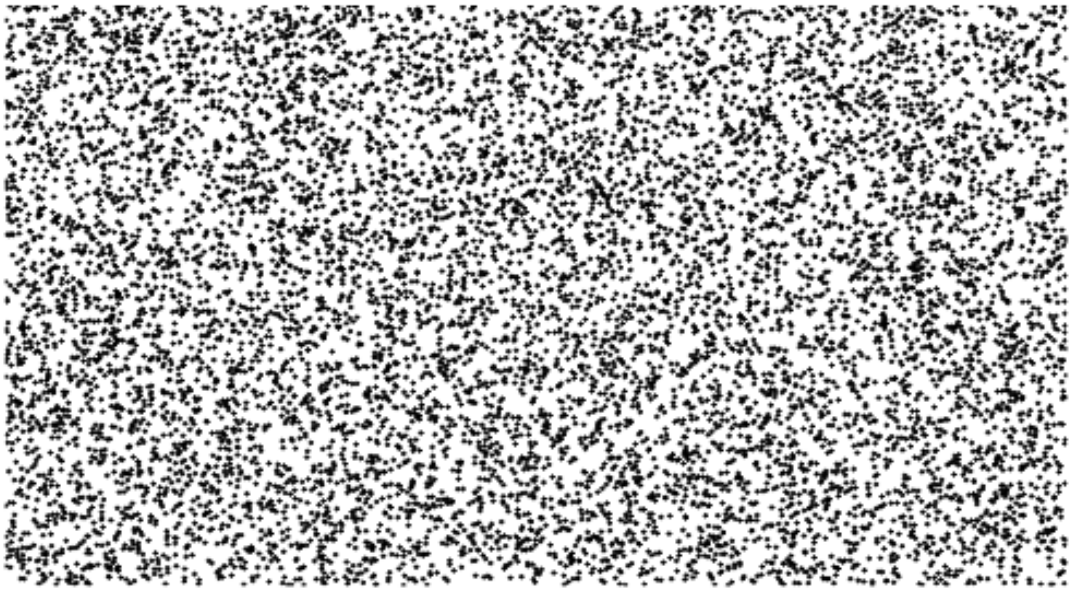
```
Loading required package: sysfonts
```

```
Loading required package: showtextdb
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.2.1     ✓ readr     2.2.0
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ lubridate 1.9.5     ✓ tibble    3.3.1
✓ purrr     1.2.2     ✓ tidyr     1.3.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

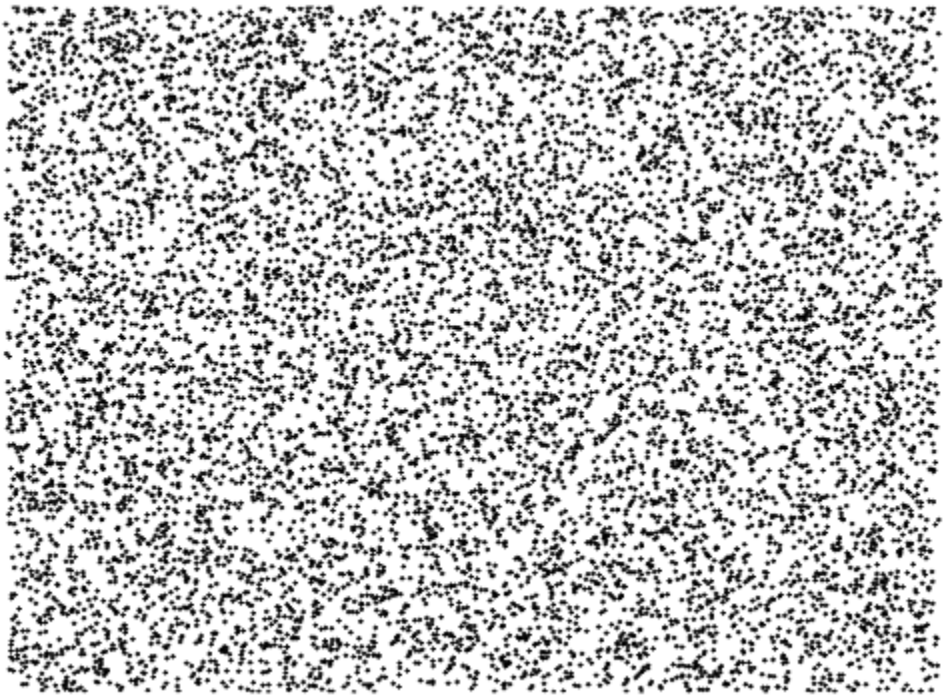
Not Sound

air

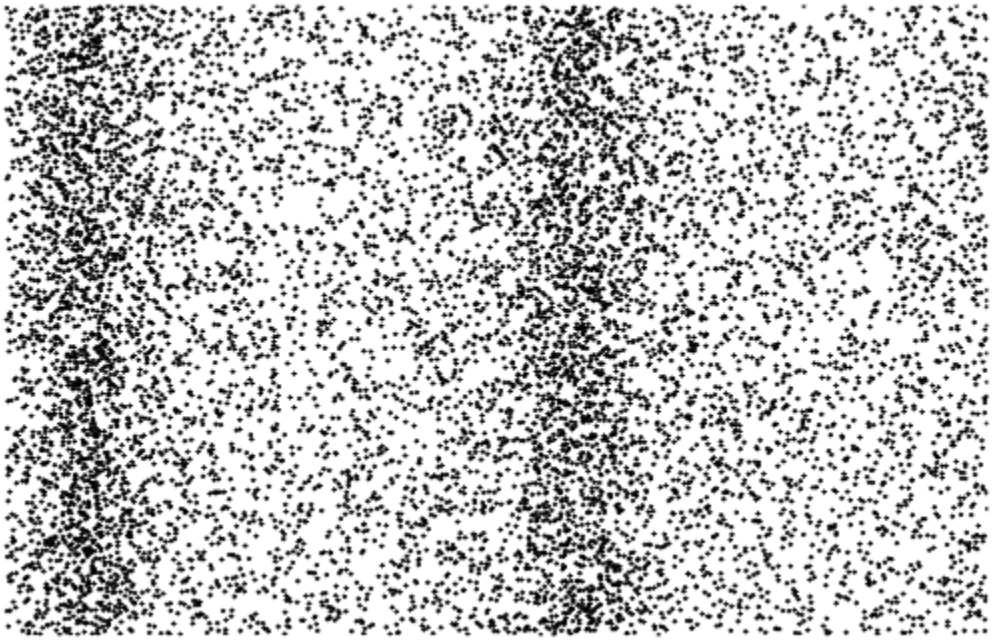




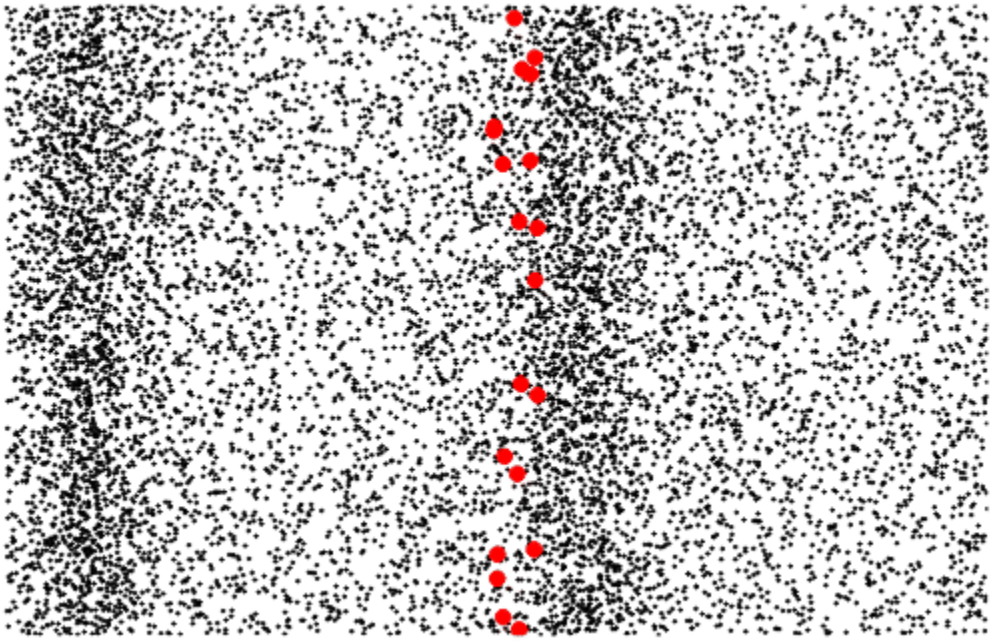
Sound - Aperiodic



Sound - Periodic

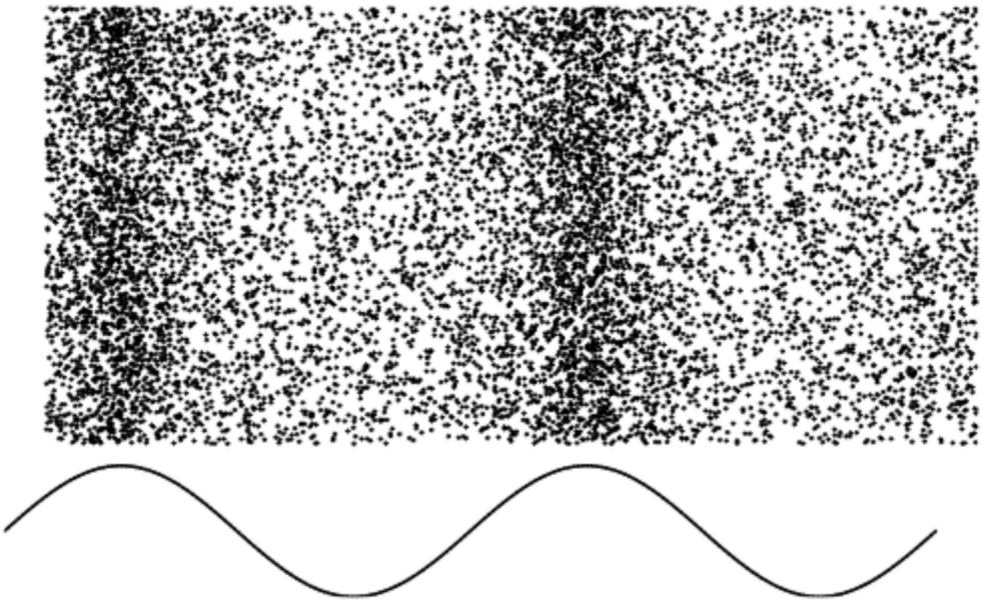


Sound



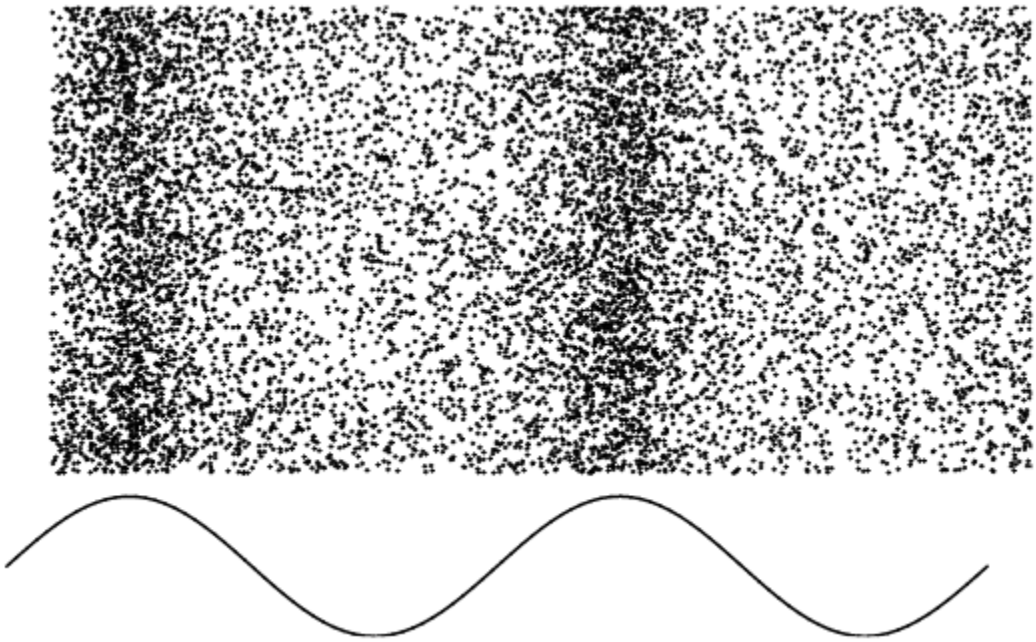
Sound is not the air molecules from my lungs hitting your ears.

Sound

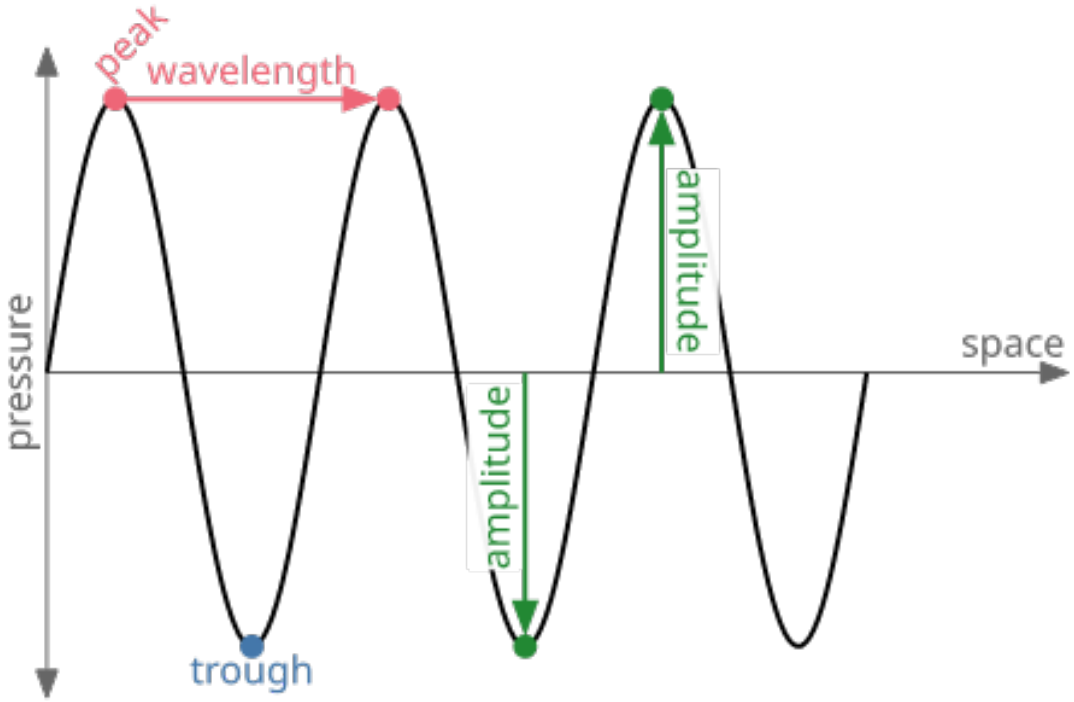


Where air is tightly packed, there's high pressure. Where it's more spread out, low pressure.

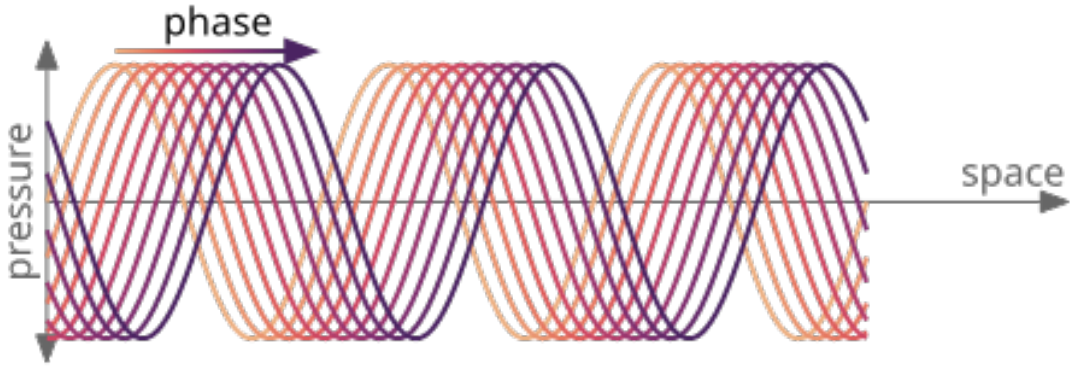
Sound



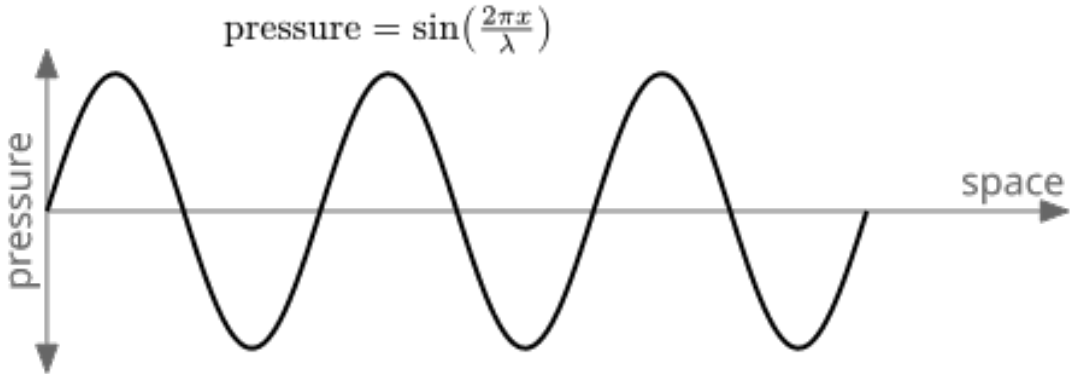
Sound


peak
wavelength
amplitude
amplitude
pressure
space
trough

Sound


phase
pressure
space

Sound - Sine Wave



Sounds - Frequency

Frequency = how many high pressure zones per second

This is going to be a function of both the wavelength and the speed of the wave.

Because the speed of sound will usually be fixed, Wavelength $\propto$ Frequency.

Frequency to Wavelength

$$\text{speed of sound} \approx 34300 \frac{cm}{s}$$

$$f \text{ Hz} = \frac{f \text{ cycle}}{s}$$

Frequency to Wavelength

$$\begin{aligned} \text{wavelength} &= \frac{34300\ cm}{s} \frac{1}{f \text{ Hz}} \\ &= \frac{34300\ cm}{s} \frac{s}{f \text{ cycle}} \\ &= \frac{34300\ cm}{f \text{ cycle}} \end{aligned}$$

Frequency to Wavelength

$$34300 \frac{cm}{s}$$

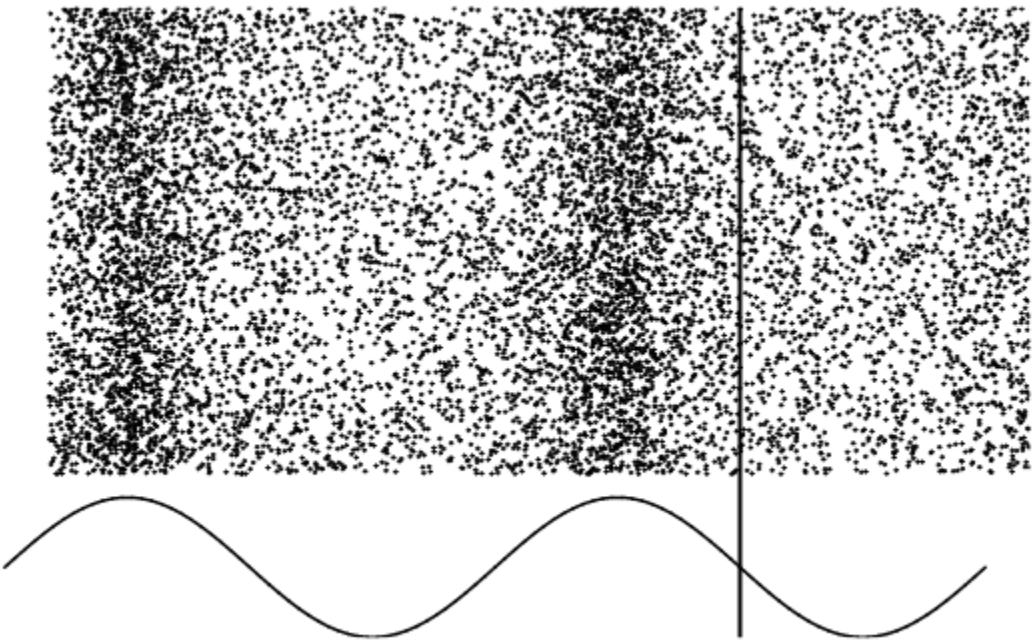
$$440 Hz = \frac{440}{s}$$

$$\frac{34300\ cm}{s}\frac{s}{440} \approx 78\ cm \approx 2.6\ ft$$

Wavelength to Frequency

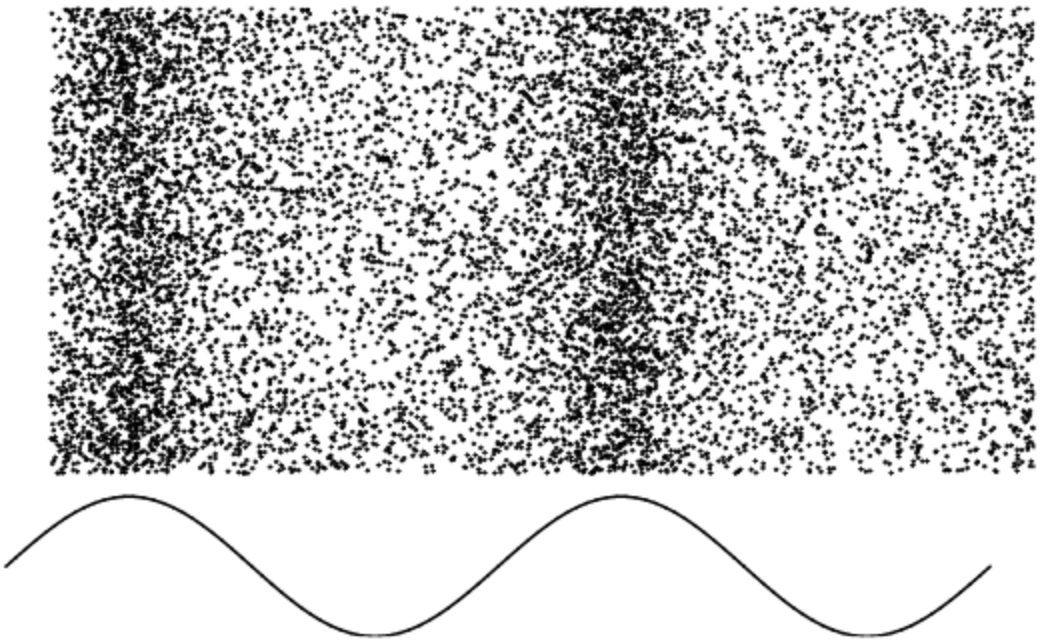
Sounds - Frequency

```
Joining with `by = join_by(t)`
```



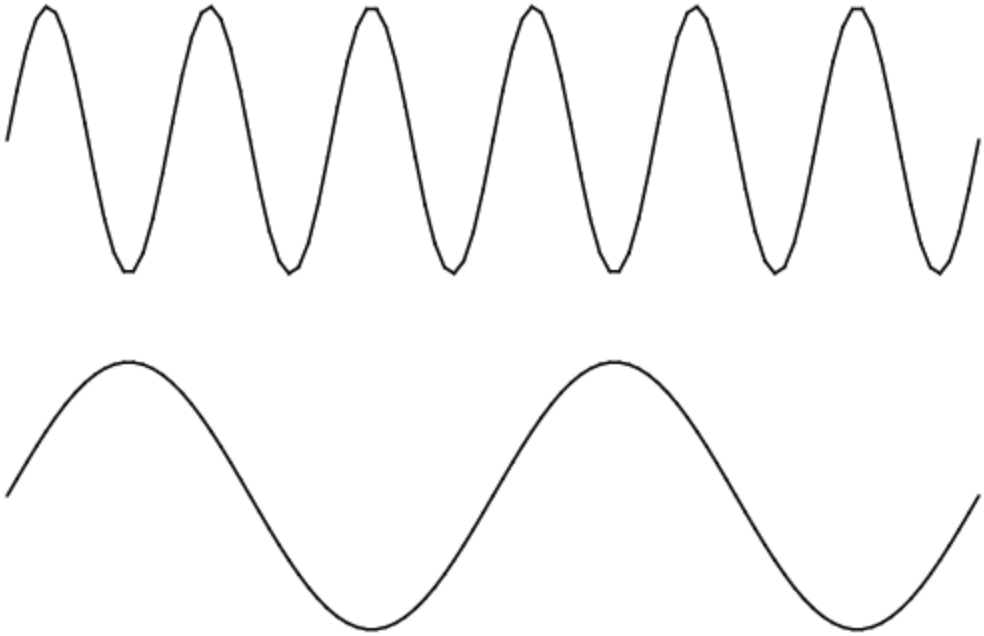
Sound - Increasing Frequency

As wavelength decreases, frequency increases



Sound - Frequency

The *speed* of these higher frequency waves is the *same* as the lower frequency waves

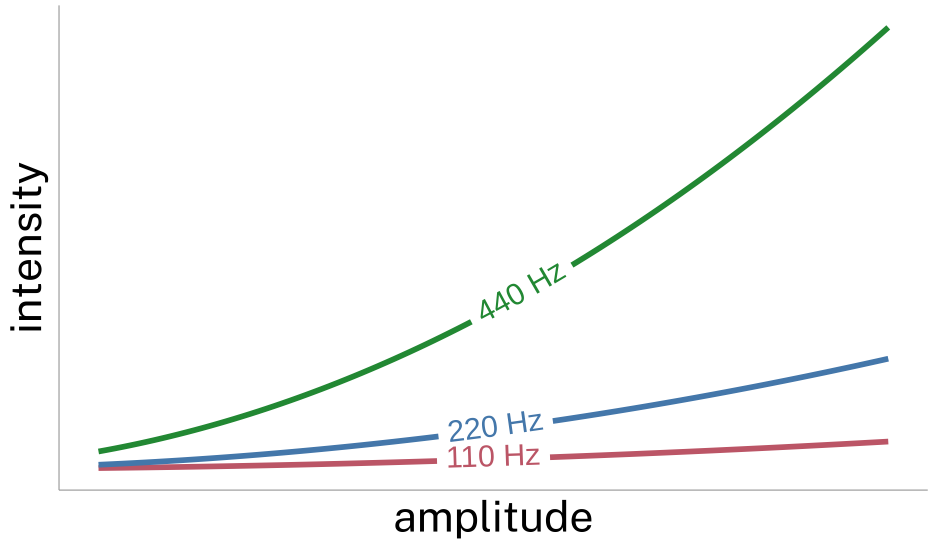


Sound - Frequency

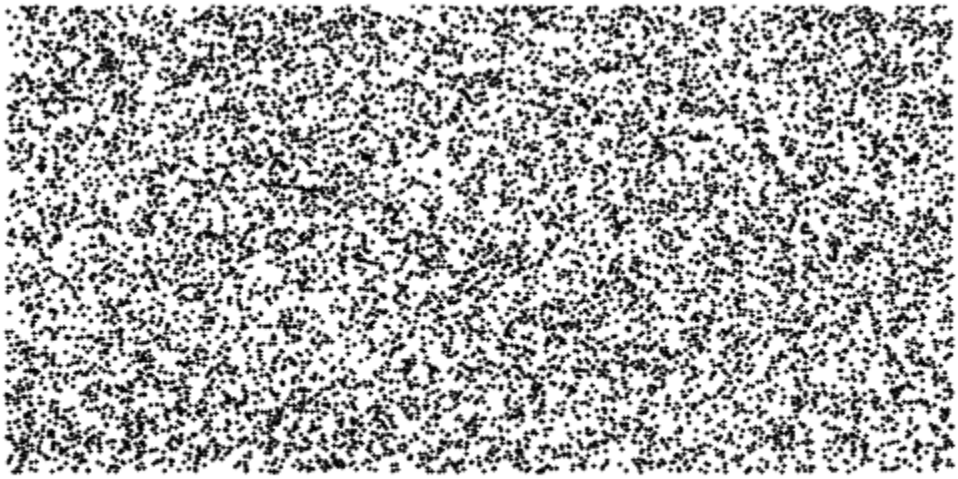
Every time frequency doubles, we experience it as the frequency increasing 1 octave.

Sound - Intensity

A relationship between frequency *and* amplitude



Sound - Increasing Amplitude



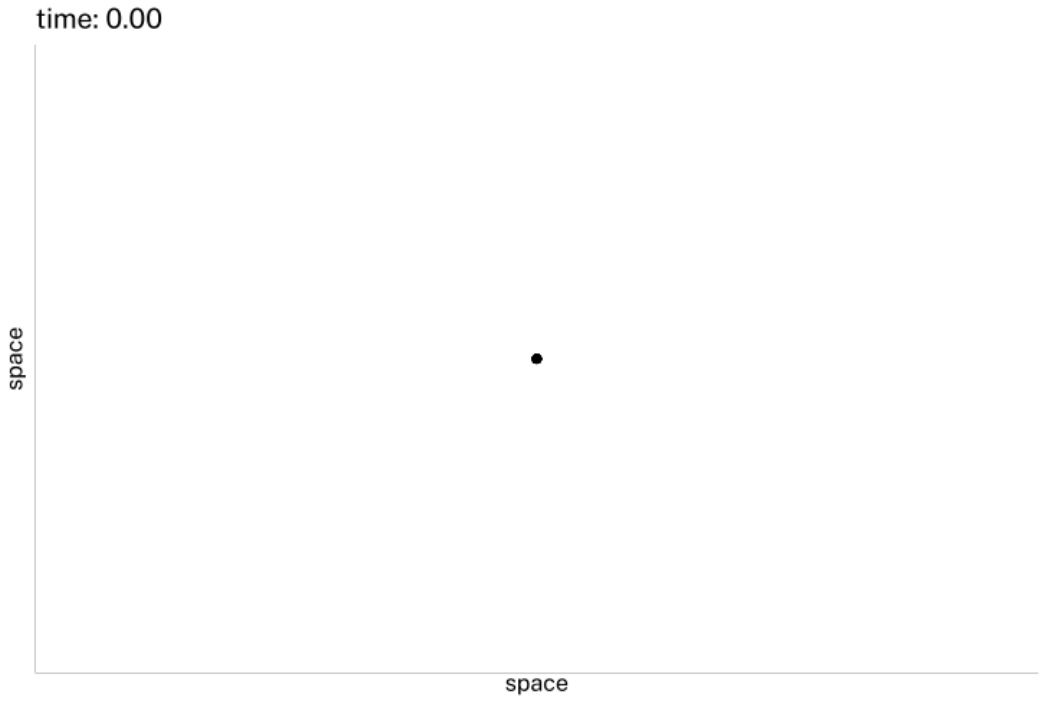
Sound - Loudness

Decibels (dB) are a measure of *relative* intensity.

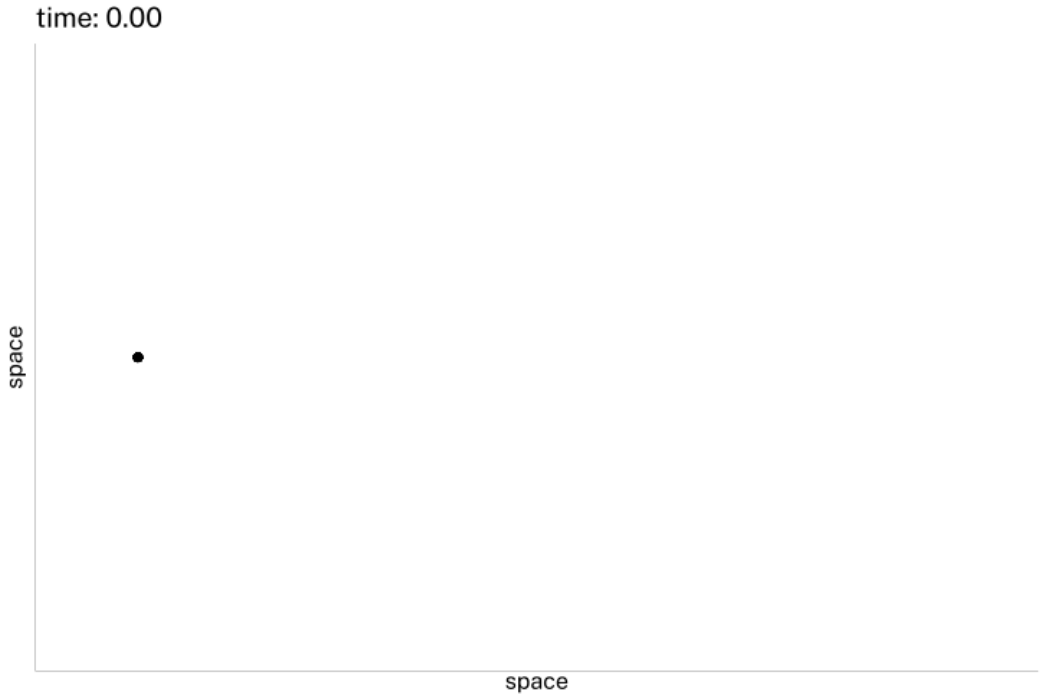
$$1dB = 10\log_{10}\left(\frac{s_1}{s_2}\right)$$



Sound - The Doppler Effect



Sound - The Doppler Effect





The Doppler Effect

```
Warning: Removed 15 rows containing missing values or values outside the scale
range
(`geom_line()`).
```

